INDUSTRIAL IT/OT INFRASTRUCTURE / MRF LIMITED, DAHEJ

Plant Network Transformation Design / Deployment / FMS Support

Voice, data and industrial connectivity engineered as one resilient foundation.



GREENFIELD BUILD TO SEVEN-YEAR OPERATIONS CONTINUITY

25.3 km	33.5 km	51	7 YEARS
SINGLE-MODE FIBRE	SHIELDED COPPER	CORE + ACCESS SWITCHES	NETWORK FMS SUPPORT

NAMED ACCOUNT: MRF LIMITED, DAHEJ / SOLUTION INTEGRATION: D-TECH

Conceptual industrial network visual. Quantities are based on supplied orders.

EXECUTIVE SUCCESS STORY

One integrated network for production, people and systems.

D-TECH designed and implemented the Dahej site foundation: industrial fibre, structured voice/data cabling, high-speed switching, racks, protected routes, testing and documentation, followed by dedicated FMS support.

49	2 × 10G	41	1 SR.
ACCESS SWITCHES	CORE SWITCHES	NETWORK / SERVER RACKS	DEDICATED FMS ENGINEER

BUSINESS OBJECTIVE

Create a dependable digital backbone for a modern plant.

Consistent connectivity across the data centre, production, offices and operational endpoints, with physical resilience, fibre reach, 10 GbE aggregation, labels and accountable support.

ENGAGEMENT MODEL

01	02	03	04
DESIGNED	SUPPLIED	DEPLOYED	SUPPORTED
Zones, backbone, racks and pathways.	Active network and passive components.	Cabling, termination, switching and testing.	Seven-year continuity through on-site FMS.

DOCUMENTED VALUE

Approximately USD 0.17 million equivalent across the supplied infrastructure and installation orders.

BUSINESS AND TECHNICAL REQUIREMENTS

Industrial-grade connectivity with operational discipline.

The programme joined production availability, IT governance, safety, maintainability and expansion in one plant-network foundation.

01 Plant-wide reach

Production, utility, warehouse and administration areas across a large campus.

02 IT/OT convergence

Controlled transport for business and industrial systems with logical separation.

03 Voice and data

Structured outlets, patching and cabinets for converged communications.

04 High-speed core

10 GbE fibre aggregation through dual data-centre-class switching.

05 Physical resilience

HDPE, PVC, flexible conduit, GI trays and appropriate routing.

06 Maintainability

Standard racks, labels, spares and as-built information.

07 Commissioning proof

Certify copper and fibre, uplinks and port-to-outlet mapping.

08 Lifecycle operations

An experienced on-site engineer plus defined escalation.

DESIGN PRINCIPLE

Treat passive infrastructure, active switching and service ownership as one operational system.

DOCUMENTED SCOPE AT A GLANCE

A large, measurable industrial network build.

The orders cover backbone, horizontal cabling, switching, cabinets, accessories, pathways, installation and handover records.

LAYER	SCALE	ROLE
Fibre backbone	25.3 km	12-core single-mode OFC across plant zones.
Shielded copper	33.5 km	67 × 500 m reels of Cat6A-class cable.
Access switching	49 units	42 standard + 7 enhanced 24-port L2 switches.
Core switching	2 units	48 × 10 GbE aggregation switches.
10 GbE optics	34 units	Long-reach single-mode SFP+ connectivity.
Racks	41 total	39 wall cabinets + 2 full-height 42U racks.
Fibre termination	50 LIUs	1,243 pigtails and 165 LC-LC patch cords.
Structured endpoints	1,404 jacks	Panels, faceplates, boxes and management.
Protected pathways	30+ km	HDPE, PVC/flexible conduit, GI and mesh trays.
Implementation	Turnkey	Laying, installation, testing and documentation.

COMMERCIAL EVIDENCE

Supply order 4300074618 and service order 2400208464, both dated 31 January 2020.

INDUSTRIAL IT/OT ARCHITECTURE

Core, backbone and access with controlled service zones.

Dual 10 GbE core switching aggregates managed plant access over single-mode fibre. Segmentation supports enterprise, voice, OT and management traffic.

E **ENTERPRISE IT**

Users, servers and corporate applications

C **CORE LAYER**

Dual 10 GbE data-room switching

F **FIBRE BACKBONE**

12-core links with LIU termination

A **ACCESS LAYER**

Managed 24-port distributed switches

I **IT / VOICE**

Structured user and phone outlets

O **OT / PLANT**

Controlled industrial endpoints

CONTROL MODEL

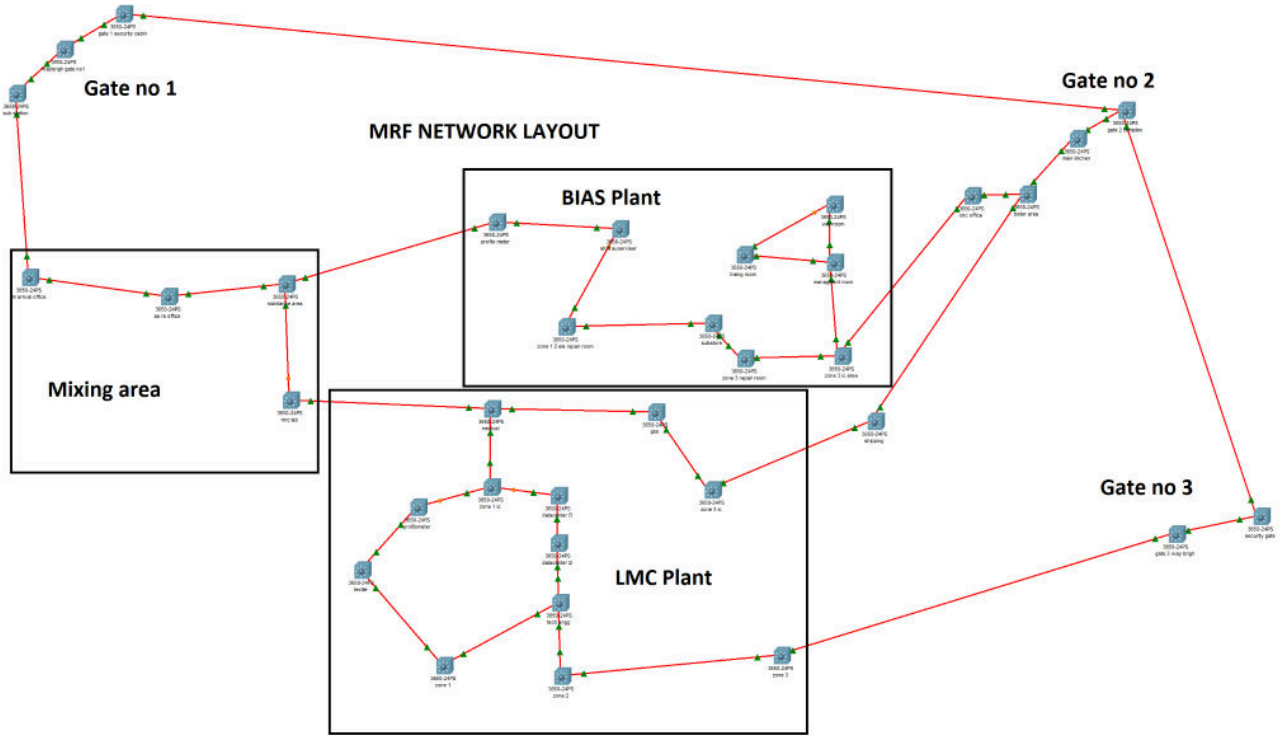
Segmentation / Managed uplinks / Monitoring / Documented ownership

Logical separation and change control support operational traffic while allowing managed integration.

ACTUAL PLANT-WIDE NETWORK TOPOLOGY

Multi-zone access across the Dahej campus.

The supplied topology maps Gate 1, Gate 2, Gate 3, Mixing, BIAS, LMC, utilities, warehouse, shipping and security onto a managed network foundation.



01

CAMPUS REACH

Three gates and production, utility and logistics zones.

02

STANDARD EDGE

Managed access switches in distributed cabinets.

03

MAPPED OWNERSHIP

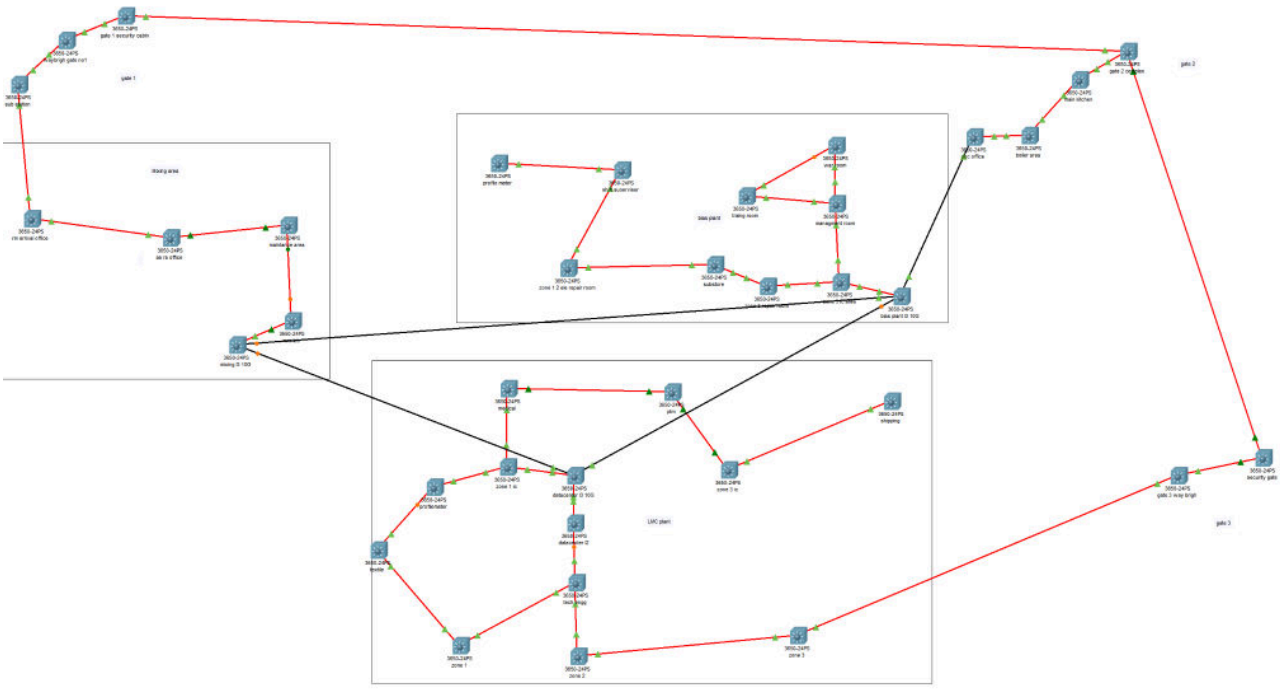
Remote locations represented as engineered nodes.

Actual project topology supplied by D-TECH; simplified for client-safe presentation.

10 GBE CORE RING ARCHITECTURE

High-speed paths connect critical plant clusters.

The documented layout links Mixing, BIAS, LMC and key locations through 10 GbE paths. An access ring extends connectivity around the campus.



MRF NETWORK LAYOUT WITH CORE SWITCH 10G RING

LARGE-SOLUTION ADVANTAGE

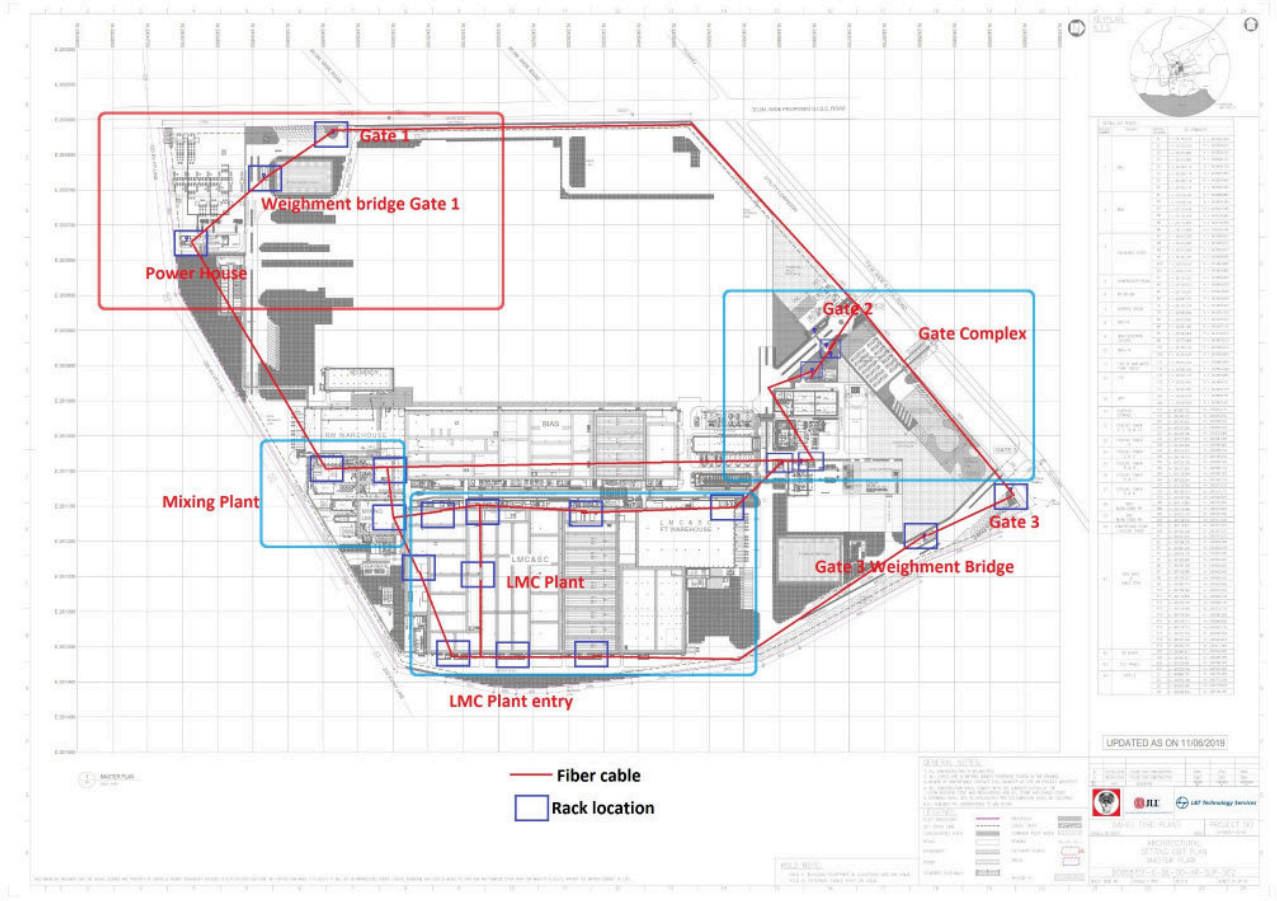
10 GbE aggregation / Alternate paths / Controlled fault domains

The layout reduces dependence on a single long chain and supports recovery and planned maintenance.

FIBRE ROUTE AND RACK MASTER PLAN

Network design aligned to the plant geography.

The master plan locates fibre routes and cabinets against buildings, gates, weighbridges, production areas and site corridors.



01

ROUTE VISIBILITY

Fibre paths tied to actual corridors.

02

RACK PLACEMENT

Cabinets shorten edge cable runs.

03

EXPANSION CONTROL

Mapped routes support planned capacity.

04

HANDOVER CLARITY

The plan becomes an operations reference.

INSTALLED NETWORK CABINET

42U core cabinet engineered for clarity and growth.

Actual site photos show fibre termination, patching, managed switching, labels, cable control and expansion space.



FULL FRONT VIEW



ANGLED SERVICE VIEW

01

STRUCTURED LAYERS

Fibre, patch panels, switches and managers follow a logical sequence.

02

TRACEABLE CABLING

Blue copper and yellow fibre labels support fault isolation.

03

EXPANSION READY

Reserved rack space and access support growth and maintenance.

Actual installed cabinet photographs supplied by D-TECH.

ACTIVE NETWORK ENGINEERING

A 10 GbE core feeding distributed plant access.

Data-centre-class switching, managed edge switches, long-reach optics, spares and support form the active-network layer.

CORE

2 × 48-port 10 GbE aggregation switches

High-density SFP+ connectivity, redundant-power design, OS10-class software and mission-critical support formed the central switching basis.

ACCESS

49 managed 24-port switches with 10 GbE uplink capability

01 Hierarchical design

Core-to-access topology simplifies operations and expansion.

02 Long-reach optics

34 single-mode 10 GbE LR modules support campus distances.

03 Operational spares

Documented spares reduce replacement delay.

04 Serviceability

Standard switch families aid configuration and troubleshooting.

05 Segmentation ready

VLAN and policy design can isolate IT, OT, voice and management.

06 Support alignment

Critical devices backed by OEM support and D-TECH escalation.

PASSIVE INFRASTRUCTURE ENGINEERING

Protected, labelled and maintainable connectivity.

The passive layer supports industrial routing, clean termination and long-term serviceability across production and administrative environments.

01

FIBRE BACKBONE

12-core single-mode OFC, LIUs, pigtails and LC patch cords.

02

COPPER ACCESS

Shielded cabling, patch panels, jacks, faceplates and boxes.

03

NETWORK CABINETS

15U wall racks and central 42U cabinets for termination and switching.

04

INDUSTRIAL PATHWAYS

HDPE/PVC conduit, flexible pipe, GI trays and mesh trays.

INSTALLATION QUALITY CONTROLS

ROUTE	Separate power and data; protect bends and crossings.
TERMINATE	Maintain fibre cleanliness, bend radius and shield continuity.
LABEL	Map rack, panel, port, outlet and destination.
TEST	Certify fibre loss and copper channel performance.
DOCUMENT	Update schedules, drawings, results and as-builts.

Good pathways and labels reduce fault-finding time, protect uptime and make growth predictable.

DEPLOYMENT AND COMMISSIONING

Controlled execution inside an operating plant.

Engineering, materials, safety, installation, configuration, validation and documentary closure were treated as connected work stages.

01 Survey and zoning

Confirm buildings, production, rack positions, routes and distances.

02 Low-level design

Prepare backbone, access, rack, port and pathway schedules.

03 Material staging

Verify switches, optics, racks, cables and spares.

04 Route construction

Install conduits, HDPE and trays with safe supports.

05 Cable deployment

Pull fibre and copper within protection and bend limits.

06 Termination and dressing

Complete LIU, patch-panel, jack and cabinet management.

07 Configuration and testing

Validate switching, uplinks, segmentation, optics and links.

08 Documentation and handover

Submit tests, labels, maps, as-builts and spares.

HANDOVER STANDARD

No link is complete until it is tested, labelled, mapped and assigned an operational owner.

SEVEN-YEAR NETWORK FMS CONTINUITY

From project completion to sustained ownership.

One senior on-site resource supports monitoring, incidents and change, with remote specialist and OEM escalation.

OPERATING MODEL

1 senior engineer / On-site ownership / Remote escalation

First point for monitoring, incidents, moves-adds-changes, documentation and coordination with plant IT, users, OEMs and D-TECH specialists.

01

DETECT

Monitor alarms and link health.

02

DIAGNOSE

Check ports, optics and configuration.

03

RESTORE

Recover service or apply controlled workaround.

04

ESCALATE

Engage specialist or OEM with evidence.

05

CLOSE

Record cause, action and prevention.

Preventive checks

Racks, switches, optics and environment.

Incident response

L1/L2 isolation, restoration and escalation.

Change support

Port changes and controlled updates.

Documentation

Asset, port, topology and change records.

Vendor coordination

OEM tickets and replacement logistics.

Service reporting

Open issues, ageing, repeats and capacity.

BUSINESS VALUE DELIVERED

A network foundation that continues to earn its value.

Performance, availability, fault isolation and managed changes matter after implementation as much as they do at handover.

01 Coverage

Fibre and copper reach supports production, offices and utilities.

02 Performance

10 GbE aggregation supports modern applications.

03 Availability

Spares, OEM support and on-site FMS aid restoration.

04 Maintainability

Standard racks, labels and maps improve fault isolation.

05 Governance

Documented changes reduce unmanaged network drift.

06 Scalability

Modular access and fibre support planned expansion.

07 IT/OT readiness

Segmentation-ready design supports controlled industrial traffic.

08 Lifecycle trust

Seven years of support extends continuity beyond handover.

SUCCESS INDICATOR

The infrastructure programme matured into a long-term managed service relationship.

EVIDENCE AND REPEATABLE BLUEPRINT

Industrial infrastructure with lifecycle accountability.

The engagement brings consulting, design, supply, deployment, testing, documentation and sustained manpower support together.

D-TECH CAPABILITY

One accountable partner across the network lifecycle

Survey and architecture / IT/OT design / Active and passive supply / Fibre and copper deployment / Rack and pathway engineering / Configuration, handover and FMS

EVIDENCE	REFERENCE	BASIS
Supply evidence	MRF PO 4300074618	31 Jan 2020
Implementation	MRF service order 2400208464	31 Jan 2020
Customer site	MRF Limited, Dahej	Manufacturing
Lifecycle continuity	Seven-year network FMS	1 senior on-site

DISCLOSURE

Commercial contacts, tax identifiers, signatures and unit pricing are excluded. USD value is a rounded presentation equivalent, not an invoice amount.

DESIGN. DEPLOY. OPERATE. IMPROVE.